Part Three

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October 12, 2026

1 Algorithmic Design and Implementation Details

Page target: 0.5 - 1 page, excluding code blocks, tables, and figures State the mathematical objective function. Derive and state the gradient equation used for parameter updates. Provide clean pseudocode for your custom gradient descent function. Detail hyperparameter settings, including learning rate , maximum iterations, and stopping criteria. Detail your software environment, dependencies (e.g., Python 3.x, NumPy, pandas, statsmodels), and exact dataset partition steps (900 train / 130 test split).

2 Feature Scaling and Gradient Descent Dynamics

Page target: 1 page, excluding code blocks, tables, and figures Summarize your approach and results of the Part 1 analyses: • Scaling and Convergence Speed • Scaling and The Optimal Solution Include any relevant figures/tables/charts.

3 Linear vs. Non-Linear Feature Transformations

Page target: 1 page, excluding code blocks, tables, and figures Summarize your approach and results of the Part 1 analysis: Non-Linear Transformations Include any relevant figures/tables/charts.

4 Univariate vs. Multivariate Model Dyanamics

Page target: 1 page, excluding code blocks, tables, and figures Summarize your approach and results of the Part 1 analysis: Univariate vs. Multivariate Coefficients Include any relevant figures/tables/charts.

5 Statistical Inference and Regression Analysis

Page target: 1 page, excluding code blocks, tables, and figures Summarize your approach and results of the Part 1 analysis: Statistical Significance (p-values)

6 Code and Reproducibility Access

Provide explicit instructions to reproduce all experimental results, including command- line arguments or notebook execution orders. Include a working link to a public GitHub repository or specify attached code files containing your implementation.